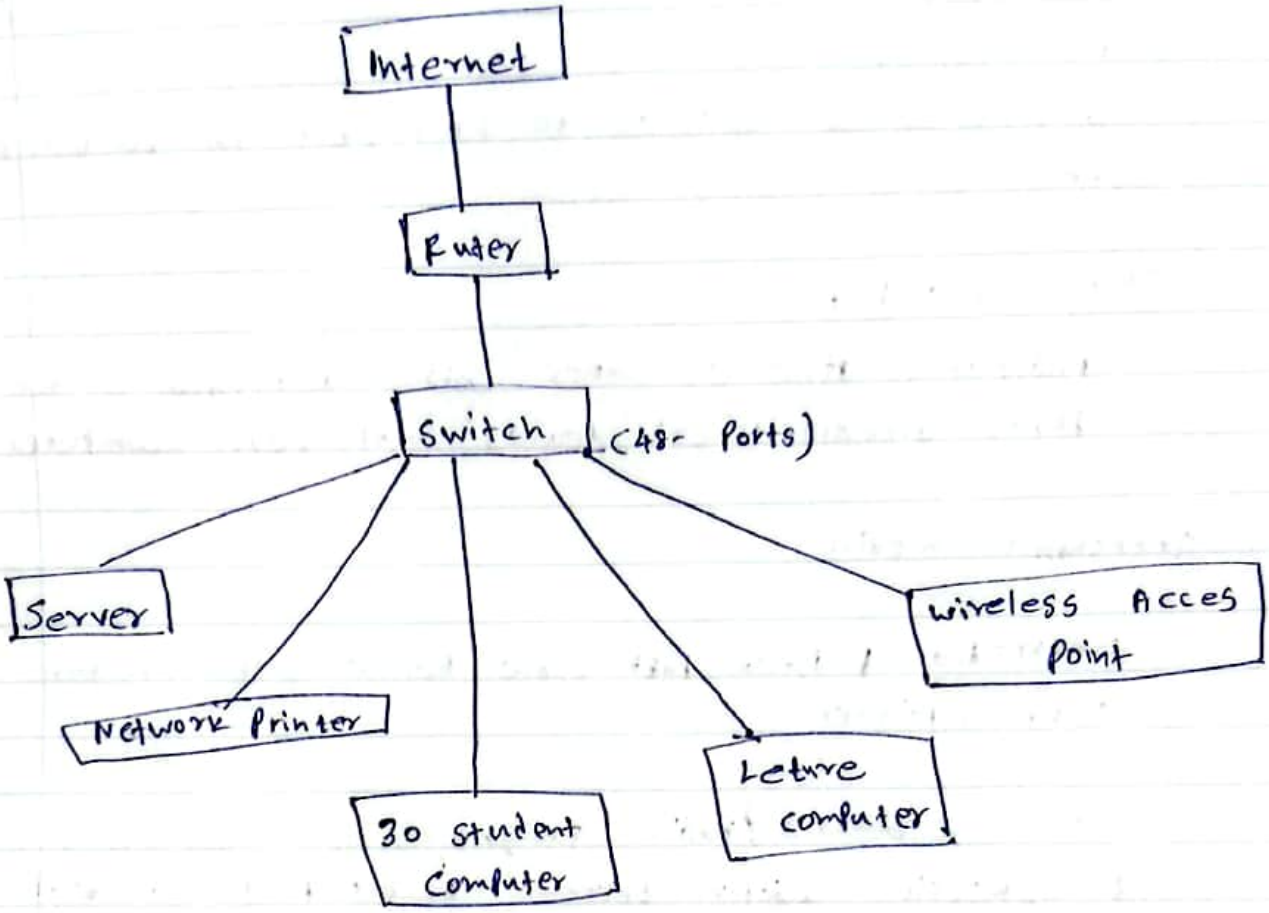


Router
4c- Port managed ethervent switch
wireless Access Point (WAP).

Why Select

Router - connects the local area network to the external wide area network
Handeles ip routing, network Address Translator to allow internal devices to share a Public ip

Swith - There is need for 34 minimum Ports and to manage LAN

Internet → router
Isp cable connects to WAN ports of router

Router → switch

ethernet cable connects to LAN port of router to switch

Switch → End devices

Individual ethernet cables connect dedicated switch ports directly to end devices, WAP, sever, computer

Transmission media

unshielded twisted pair cable between switch, router server, printer

wireless signals / radio frequency to wirless transmission medium between access point and mobile devices

when student reguest a webpage, browser encapsulates the reguest into data packets containig the source ip and destination ip
the nic transmit that signal to switch. switch reads the MAC address and forward into the router after router recives data packets route performs NAT to swap LANIP with router's public WAN when web sever responds router takes incoming data and performs NAT and forward to switch, then based on MAC address switch forward data